

This flow uses the open source software CloudCompare.

### 1. Import the Point Cloud in CloudCompare

Load your point cloud (.LAS, .LAZ, .PLY, .XYZ, .PTS, .E57, etc.)

### 2. Clean the Point Cloud

Before meshing, remove unwanted points, remove obvious outliers.

Select the cloud

Edit → Scalar Fields → Filter by Value

(if intensity or confidence exists)

or

Tools → Clean → SOR filter (Statistical Outlier Removal)

### 3. Crop Unwanted Areas

Edit → Segment

Draw a polygon around the object, keep the selected region. Delete the rest.  
Repeat until only the desired object remains.

### 4. (Optional) Downsample

Very dense clouds slow meshing. Edit → Subsample

### 5. Estimate Normals

Meshing algorithms need surface normals. Select the cloud.

Edit → Normals → Compute

### 6. Generate the Mesh

Option A: Poisson Surface Reconstruction (Best for closed objects)

Go to: Plugins → PoissonRecon

Pros: Smooth, Fills holes automatically, Good for statues, rocks, scanned objects

Cons: Creates surfaces where data is missing.

Option B: Delaunay 2.5D Mesh

Go to: Tools → Projection → Delaunay 2.5D

Best for: Terrain, Roads, Roofs, Flat landscapes

Not suitable for vertical walls or complex 3D objects.

## 7. Clean the Mesh

After reconstruction, remove floating triangles. Delete isolated components.  
Smooth if needed.

Edit → Mesh → Smooth

Edit → Mesh → Remove Small Components

## 8. Check Mesh Quality

Inspect for: Holes, Non-manifold edges, Flipped normals, Spikes, Noise

If necessary: Recompute normals.

Increase/decrease Poisson depth.

Improve point cloud cleaning.

## 9. Export the Mesh

File → Save